

Chapter 73 — Priapism

- **Priapism** is a penile erection that is either **unrelated to sexual stimulation or persists ≥ 4 hours** after its conclusion; it results from dysregulation of the normal processes governing rigidity and detumescence, and it is a **medical emergency**.
- The **first step is to characterise penile haemodynamics and oxygenation**; appropriate treatment follows from that characterisation.
- **Guidelines** to be familiar with: American Urological Association (AUA)/Sexual Medicine Society of North America (SMSNA) (acute ischaemic priapism 2021; recurrent ischaemic, sickle cell and nonischaemic priapism 2022), European Association of Urology (EAU) (2014), and British Association of Urological Surgeons (BAUS).
- **Time is of the essence** because of the high potential for adverse sequelae: pain, infection, fibrosis, penile shortening and permanent erectile dysfunction (ED).
- Many aspects of the disease process **still require investigation**.

PRIAPISM: DEFINITION AND CLASSIFICATIONS

Priapism Definition

- **Full or partial erection ≥ 4 hours** after sexual stimulation and orgasm, or unrelated to sexual stimulation.
- **Three subtypes**: ischaemic (low flow, veno-occlusive), nonischaemic (high flow, arterial) and stuttering (intermittent, recurrent ischaemic).

Priapism Classifications/Subtypes

Ischaemic Priapism (Low Flow, Veno-Occlusive)

- Painful erection with **low inflow of oxygenated arterial blood**; with time, tissue develops **hypoxia, hypercarbia and acidosis**.
- **Penile pain develops within 6–8 hours** of onset and usually prompts emergency presentation.
- Examination: **fully erect, rigid, tender**, sometimes oedematous.
- **Progressive; likened to compartment syndrome**: high intracavernosal pressure, sinusoidal thrombosis, diminished arterial inflow and tissue ischaemia.
- **Untreated, it takes days to weeks to resolve** and commonly ends in penile fibrosis and permanent loss of erectile function.

Stuttering Priapism (Intermittent, Recurrent Ischaemic Priapism)

- **First described in sickle cell disease (SCD)**.
- **Painful, recurrent, unwanted erections** lasting minutes to hours; **often nocturnal**, waking the patient.
- **Onset usually in childhood**; over time episodes become **more frequent and longer**.
- Episodes usually self-resolve, but **can progress to a full ischaemic episode** requiring treatment.

Nonischaemic Priapism (High Flow, Arterial)

- **Dysregulated arterial inflow**; corpora erect but **usually lacking full rigidity**.
- Arterial blood with **normal oxygen tension** → **no pain, no hypoxia, hypercarbia or acidosis** → fibrosis and ED usually do not follow.
- Once confirmed, **no acute intervention is needed**; expectant management may be pursued.

- Usually follows **trauma** (blunt perineal trauma, iatrogenic penile needle injury) → **cavernosal artery–corpus cavernosum fistula** → unopposed arterial inflow → sustained erection.

THE THREE SUBTYPES COMPARED

	Ischaemic	Stuttering	Nonischaemic
Lesion	Veno-occlusion, failed outflow	Recurrent ischaemic episodes	Arterial–sinusoidal fistula
Share of cases	95% (commonest)	Subset of ischaemic	<5% (rare)
Pain	Painful	Painful, intermittent	Painless
Rigidity	Full	Full during episode	Partial
Oxygenation	Hypoxic, hypercarbic, acidotic	Ischaemic per episode	Normal (arterial)
Typical cause	Drugs, SCD, haematological	SCD	Trauma
Urgency	Emergency	Emergency per episode + prophylaxis	Not an emergency
Fibrosis/ED	Common if untreated	Risk with each episode	Usually absent

- The table tests **one axis — oxygenation**; pain, rigidity, urgency and prognosis all follow from it. **Trap**: stuttering priapism is ischaemic, not a separate benign entity.

KEY POINTS: DEFINITION OF PRIAPISM

- Priapism is a penile erection that is either not related to or continues for ≥ 4 hours after the conclusion of sexual stimulation.
- Priapism is typically classified as ischaemic (low flow, veno-occlusive), stuttering (intermittent, recurrent ischaemic) or nonischaemic (high flow, arterial).

Concept — Priapism is classified by oxygenation, not by duration: **failed venous outflow produces ischaemia and a time-dependent emergency; excess arterial inflow produces a well-oxygenated, painless erection that can wait.**

PRIAPISM HISTORICAL CONSIDERATIONS

- **Priapus**: Greek god depicted with an erect penis and strong sexual desire; son of Zeus and Aphrodite; **cursed by Hera with abnormally large genitals**; rejected by Aphrodite and raised by shepherds; wherever he went flowers bloomed and animals mated — "**god of fertility**".
- **Nineteenth-century reports established conservative therapy as the norm**; the turn towards surgery began with Morris.
- **Nonischaemic priapism**: literature is small series and case reports, focusing on **observation as first-line**, then embolisation for refractory cases.
- **Embolisation agent and ED**: historically temporary agents were thought to carry lower long-term ED; more recent series report **lower long-term ED with permanent agents**.

HISTORICAL MILESTONES

Year	Figure	Distinguishing contribution
1824	Callaway	First reported case ; 16 days of conservative care including leeches failed → lancet puncture of left crus drained "grumous blood" → lifelong impotence
1845	Tripe	"Stout seaman" treated conservatively (rhubarb, cold lotions, tartar emetic, calomel, colocynth) → "perfectly cured"; argued against instrumental cure
1873	Hird	Benefit from bromide and potassium; incisions of no benefit

Year	Figure	Distinguishing contribution
1895	Henry Morris	Medical treatments "useless"; incisions deserve further trial — reopened surgery
1914	Frank Hinman Sr	Medical therapy futile; classification "nervous" vs "mechanical" ; aseptic corporal incision and drainage ; described painful "transitory erections" distinct from persistent ones
1960	Frank Hinman Jr	Seven iatrogenic cases; light microscopy : trabecular oedema → fibrosis; venous stasis with ↑CO₂ and ↓O₂ tension ; venous drainage must be applied early
1960	Burt et al.	First report of nonischaemic priapism
1980	Emond et al.	Jamaican SCD cohort; coined "stuttering priapism"

- Pattern: **conservative** → **surgical** → **mechanistic**. Trap: Hinman Sr (1914, classification and aseptic drainage) vs Hinman Jr (1960, histology and venous stasis).

KEY POINTS: HISTORICAL CONSIDERATIONS

- The term priapism is derived from the name of the Greek god Priapus, who was typically depicted as having an erect penis.
- The first report of priapism in the medical literature was published in 1824.

Concept — *The history swings between conservative and invasive treatment until Hinman Jr identifies venous stasis with hypoxia and hypercapnia as the mechanism, making early drainage the logical treatment.*

THE EPIDEMIOLOGY OF PRIAPISM

- **Population incidence 0.34–5.3 cases per 100,000 person-years** across registries and national data sets.
- **Reported rates rose sharply with intracavernosal injection (ICI) therapy** for ED — the Finnish registry doubled in its final years, coinciding with ICI introduction.
- **SCD is a major risk factor** for ischaemic and stuttering priapism: **29–42% of males with SCD** develop priapism in their lifetime. **Population SCD prevalence drives population priapism incidence.**
- Higher incidence in **older men (aged 40 and over)** in Dutch general-practice data; United States (US) inpatient data show a **peak at 40–44 years**, mean age early thirties, **predominantly Black** patients.
- **Inpatient data sets undercount**: only a **small minority of emergency department visits are admitted**, so emergency-department-based data give the highest incidence; **quoting inpatient data as true incidence is a trap.**
- **Seasonality**: incidence higher in **summer** than winter.
- **Predictors of admission**: Charlson comorbidity index ≥3, Medicaid (vs private) insurance, very high emergency department volume, **SCD** and **drug abuse** (increase); rural hospital and very high ZIP-code income (decrease).
- **SCD share is highest in children** and in admitted patients.
- **Readmission**: about a quarter readmitted within a year, **mostly within 60 days**; most only once. Risk factors: SCD, drug abuse, psychiatric disease, prior ED, non-commercial insurance. **Inpatient admission at first presentation was protective.** Identifying these factors allows intervention to prevent readmission.

KEY POINTS: EPIDEMIOLOGY OF PRIAPISM

- Population-based estimates of priapism incidence range from 0.34 to 5.3 cases per 100,000 person-years.

- SCD is a major risk factor for priapism.
- The incidence of priapism within a population is closely linked to the prevalence of SCD within that population.

Concept — Incidence tracks **two exposures** — **ICI therapy and SCD prevalence** — and is underestimated by any data set that counts only admissions.

THE AETIOLOGY OF PRIAPISM

Aetiology of Ischaemic Priapism (Low Flow, Veno-Occlusive)

- **95% of all priapism**; causes numerous and wide-ranging.
- Erection **starts normally**; beyond 4 hours → ↓ O₂, ↑ CO₂, acidosis → **cavernosal thrombosis and smooth muscle necrosis**.
- Natural course untreated: painful erection lasting days to weeks → **fibrosis and ED**.

Intracavernosal injection therapy

- ICI agents: **papaverine, phentolamine, prostaglandin E1 (PGE1)**; prescribed after an **in-office test dose** to set a safe starting dose.
- **Low priapism risk at clinic initiation** of PGE1.
- At home: **incorrect dosing, deliberate escalation, or recreational use**.
- **Recreational ICI** (without prescription) accounted for about half of ICI-related ischaemic priapism in a Los Angeles series; recreational users were **younger, presented later**, and a **high proportion were HIV-positive** — a public-health issue (drug "from a friend": no dosing, no counselling, possible needle sharing).

TABLE 73.1 MEDICATIONS ASSOCIATED WITH ISCHAEMIC PRIAPISM

Drug class	Medications	Mechanistic grouping
α-Adrenergic receptor antagonists	Doxazosin, prazosin, tamsulosin, terazosin	Loss of α-mediated detumescence
Antidepressants and antipsychotics	Trazodone , bupropion, chlorpromazine, clozapine, fluoxetine, lithium, olanzapine, phenothiazines, risperidone, sertraline, thioridazine	Many carry α-antagonist activity
Antihypertensives	Guanethidine, hydralazine, propranolol	Vasodilatation / sympathetic blockade
Vasoactive erectile agents	Injected: papaverine, phentolamine, PGE1. Oral: avanafil, sildenafil, tadalafil, vardenafil	Direct smooth muscle relaxation
Anticoagulants	Heparin, warfarin	Heparin-dependent antibodies; rebound on cessation
Attention-deficit/hyperactivity disorder agents	Atomoxetine, methylphenidate	—
Hormonal agents	Gonadotropin-releasing hormone (GnRH), testosterone	—
Antianxiety agent	Hydroxyzine	—
Recreational drugs	Alcohol, cocaine, marijuana	—

- The table exists to drive the **medication history** in every presentation. Pattern: most agents either **block α-mediated contraction** or **directly relax smooth muscle**.

- **Trap:** oral phosphodiesterase type 5 (PDE5) inhibitors appear here as a cause, yet are used as **prophylaxis** for stuttering priapism.

Toxins

- **Tityus serrulatus** (South American scorpion) stings and **Phoneutria nigriventer** (Brazilian banana spider) bites.
- Phoneutria neurotoxin → pain, vascular congestion, priapism, loss of muscle control, paralysis, death; its peptide **eretina interferes with the nitric oxide pathway** and induces erection in mice **even after cavernous nerve denervation**.

Haematological and prothrombotic states

- **Haematological:** SCD, thalassaemia, granulocytic, myeloid and lymphocytic leukaemia, multiple myeloma.
- **Prothrombotic:** asplenia, erythropoietin, **cessation of warfarin**.
- **Haemodialysis with heparin:** proposed **heparin-dependent antiplatelet antibodies** → platelet aggregation → cavernosal flow disruption; **high haematocrit and hypovolaemia** increase risk on dialysis.

Post-shunt thrombosis and perioperative anticoagulation

- **Recurrence after a shunt may be thromboembolic**, like thrombus in a vascular graft.
- Cavernosal blood normally **does not clot** (endothelial fibrinolytic and anticoagulant factors).
- Mechanism: shunt injures tunica and endothelium → blood exposed to **collagen-rich tunica** → **tissue factor binds factor VIIa** → coagulation cascade; exposed collagen releases inorganic phosphate → thrombin activation → platelets and fibrin → **clot seals the tunica and occludes the shunt**.
- Regimen: **preoperative aspirin 325 mg orally + heparin 5000 units subcutaneously** → **postoperative aspirin 81 mg daily for 2 weeks**, to keep the cavernosum-spongiosum shunt patent.

Leukaemia

- **1–5% of males with leukaemia**; associated with high white blood cell (WBC) counts.
- **Four mechanisms:** (1) venous congestion from **splenomegaly compressing abdominal veins**; (2) **leukaemic cell sludging** in corpora and dorsal veins; (3) **sacral nerve infiltration**; (4) **central nervous system infiltration**.
- Treat **both the penis and the malignancy**: cytotoxic therapy and **leukapheresis**.

Solid tumour metastasis

- **Testicular, renal and lung** primaries can metastasise to the penis.
- Usually **obstruct venous outflow** → **ischaemic**; less commonly **destroy sinusoidal architecture** → **high flow**.
- **Individualised:** expectant management, partial penectomy, radiotherapy and/or chemotherapy; too few cases for best-practice guidance.

Sickle cell disease

- Point mutation in the **β-globin chain** producing haemoglobin S (HbS): **valine substituted for glutamic acid**.
- **Sickled cells obstruct penile venous outflow** → stagnation in corpora.
- **SCD causes 23% of adult and 63% of paediatric ischaemic priapism**.
- Multicentre study, United Kingdom and Nigeria: **mean onset about 15 years**; **most have had an episode by age 20**; distinguished **acute severe prolonged** (>24 hours, needing medical attention) from **stuttering** (short, self-limited), which was the **more common** form — typically a few episodes a month lasting around an hour.

- **Precipitants:** sexual arousal, intercourse, fever, **sleep, cold weather, dehydration**. **Self-measures:** analgesics, oral water, exercise.
- Laboratory picture: ↓ haemoglobin, ↑ reticulocytes, lactate dehydrogenase (LDH), bilirubin and aspartate aminotransferase (AST); **haemolysis and reduced nitric oxide** are central to pathogenesis.
- **Sickle cell trait:** generally benign, **low risk**; rare case reports.

Aetiology of Stuttering Priapism (Recurrent Ischaemic Priapism)

- **Most commonly in males with SCD.** About **300,000 infants** born annually with SCD, mostly in **sub-Saharan Africa, India, the Mediterranean and the Middle East**; about 100,000 affected in the US.
- Pathophysiology: deoxygenation in capillary beds → **HbS polymerisation** → sickling → ↑ adhesion molecules → abnormal endothelial binding → **rapid haemolysis** → inflammation, endothelial dysfunction, reticulocytosis → **vaso-occlusion** → acute and chronic organ damage (stroke, pulmonary hypertension, leg ulcers).
- In the penis: vaso-occlusion → **impaired cavernosal outflow** → venous congestion → impaired arterial inflow → priapism.
- **Supportive measures** (oral fluids, warm showers, walking) help some episodes.
- Episodes lasting **4–6 hours or longer** are full ischaemic priapism → seek care → **same first-line treatment as ischaemic priapism**.
- **Intravenous (IV) fluids and analgesia** may run concurrently, but **only after first-line priapism treatment is prioritised**.

Aetiology of Nonischaemic Priapism (High Flow, Arterial)

- Persistent erection from **unregulated arterial inflow**, usually after **blunt or penetrating penile/perineal trauma** → laceration of the **cavernosal artery or a branch**.
- Mechanisms: **straddle injury**, coital trauma, pelvic fracture, iatrogenic needle injury, neoplastic vascular erosion.
- **Delayed onset (≥1 day after injury) is common:** the clot occluding the arterial injury disperses, typically **during a nocturnal erection**, unmasking the fistula.
- **Iatrogenic** after treatment of ischaemic priapism by needle aspiration, irrigation or phenylephrine; colour duplex Doppler ultrasonography (DUS) shows the **fistula at the needle site**.
- Erection **partially rigid, not painful**.
- **Mimic: postischaemic hyperaemia** — common, transient after treatment of ischaemic priapism; **colour DUS distinguishes** the two.

KEY POINTS: AETIOLOGY OF PRIAPISM

- Ischaemic priapism aetiologies include exposure to certain medications and venomous toxins, as well as leukocytosis, malignancy and blood dyscrasias.
- SCD is the most common cause of stuttering priapism.
- Sickling of red blood cells can cause penile vasculature occlusion and predispose to recurrent episodes of ischaemic priapism.
- Nonischaemic priapism can be caused by trauma, with formation of a cavernosal artery–corpus cavernosum fistula.

Concept — Every ischaemic cause converges on **outflow obstruction or failed smooth-muscle contraction**; every nonischaemic cause converges on **a single mechanical lesion — an arterial–sinusoidal fistula**.

MOLECULAR MECHANISMS OF ISCHAEMIC AND INTERMITTENT PRIAPISM

- Interplay of corporal smooth muscle, endothelium and signalling through **nitric oxide (NO)**, **PDE5**, **cyclic guanosine monophosphate (cGMP)** and **transforming growth factor (TGF)- β** .
- Core defect: **imbalance between vasoconstriction and vasorelaxation** → hypoxia, hypercapnia, acidosis.
- Prolonged hypoxia → **smooth muscle apoptosis** → poor contractility and fibrosis, mediated by **reactive oxygen species (reperfusion injury)** and **TGF- β upregulation**.

Nitric Oxide Dysregulation

- Normal: NO from endothelium and cavernous nerves → **guanylyl cyclase** → **$\uparrow$ cGMP** → smooth muscle relaxation.
- **Paradox**: priapism is a **low-NO state** from dysregulated **endothelial nitric oxide synthase (eNOS)**.
- **eNOS-knockout mice show phasic priapic activity**; eNOS gene transfer partially restores erectile function.

PDE5/Cyclic GMP Dysregulation

- eNOS dysregulation acts **downstream through PDE5**: eNOS-deficient mice have **lower PDE5 and cGMP**; adenoviral eNOS restores cGMP and **abolishes priapic activity**.
- Rat cavernous cells: **ischaemia downregulates PDE5**; **sildenafil upregulates it**. PDE5 is central in sickle cell priapism.
- Mechanism: chronic low NO → altered cGMP-dependent feedback → **PDE5 downregulated** → neuronal NO surge → cGMP surge **insufficiently degraded** → unopposed smooth muscle relaxation → priapism.
- This explains the **paradoxical role of PDE5 inhibitors in preventing recurrent ischaemic priapism**.

Other Pathways: Rho, Adenosine Dysregulation

- **Rho-kinase** mediates contractility that **maintains flaccidity**; in SCD mice nerve stimulation produced a **decline in Rho-kinase activity** → unopposed relaxation.
- **Adenosine**, a pro-erectile vasodilator, is **upregulated in sickle cell priapism** in a **hypoxia-inducible factor (HIF)-1 α -dependent** manner.
- **Net effect**: NO imbalance, PDE5 dysregulation, adenosine excess and reduced Rho-kinase → **excessive relaxation with impaired vasoconstriction**.
- **TGF- β -mediated corporal fibrosis is the common downstream pathway**: $\uparrow$ TGF- β with histological change after prolonged pharmacological erection (canine); recombinant TGF- β injection reduced smooth muscle (rabbit); **TGF- β receptor gene polymorphisms predispose men with SCD to priapism**.

MOLECULAR PATHWAYS COMPARED

Pathway	Normal role	Change in priapism	Consequence
eNOS/NO	Pro-erectile via cGMP	$\downarrow$ basal NO	Loss of feedback on PDE5
PDE5	Degrades cGMP	$\downarrow$ expression	cGMP surge unopposed
Rho-kinase	Maintains flaccidity	$\downarrow$ activity	Lost contraction
Adenosine	Vasodilator	$\uparrow$ (HIF-1 α)	Excess relaxation
TGF- β	Fibrosis regulator	$\uparrow$	Fibrosis, ED

- Pattern: every change **removes a brake on relaxation**. **Trap**: assuming priapism is a high-NO state because NO is erectogenic.

KEY POINTS: MOLECULAR MECHANISMS OF ISCHAEMIC AND INTERMITTENT PRIAPISM

- Ischaemic priapism is characterised by dysregulated homeostasis and an imbalance between penile smooth muscle constriction and relaxation.
- Ischaemic priapism commonly results in acidosis, hypoxia and hypercapnia within the corpus cavernosum.
- Dysregulation of NO, eNOS, PDE5, cyclic GMP, Rho-kinase and adenosine has been implicated in the pathogenesis of priapism.

Concept — Priapism is a disorder of **failed detumescence, not excess erection**; low basal NO lowers PDE5, so the next erection cannot be switched off — which is why a PDE5 inhibitor can prevent it.

PRIAPISM: DIAGNOSIS

- Built on **history, physical examination, laboratory testing and adjunct imaging**.

History

- **Comorbidities**: haematological dyscrasias, genitourinary or systemic infection, metabolic conditions, malignancy (haematological, metastatic, primary), neurological disease.
- **Complete medication history** including hormonally active agents and recreational drugs (Table 73.1).
- **Inciting events**: penile/perineal trauma (e.g. straddle) → nonischaemic; penile/perineal surgery or pelvic vascular procedures; **recent oral or intracavernosal erectogenic agent → ischaemic highly likely**.
- Prior episodes and treatments; **duration, pain, continuous vs intermittent** rigidity.
- **Pain → ischaemic; painless → nonischaemic; intermittent pain and rigidity → stuttering**.
- **Baseline erectile function** informs counselling, particularly on **early prosthesis**; validated questionnaires are impractical acutely, but baseline function and prior therapies should be documented.

Physical Examination

- **Vital signs**: hypertension, tachycardia or bradycardia — baseline before **sympathomimetic** administration.
- General appearance and distress; expedited **abdominal, pelvic and perineal** examination for trauma, malignancy or instrumentation.
- Penis: needle marks, trauma, anomalies; **fully rigid and tender (ischaemic) vs partially rigid and nontender (nonischaemic)**.
- **Glans and corpus spongiosum are uninvolved in both**; rigidity or abnormality there suggests **alternative pathology**.

Laboratory Testing**Corporal Aspiration and Blood Gas**

- **Cornerstone and definitive test** for ischaemic vs nonischaemic; obtain in most cases. **Exception: known recent ICI of vasoactive medication**.
- Even after recent ischaemic priapism, repeat gas to assess acidosis or **conversion to nonischaemic**.
- **Aspirate colour is the first indicator**: dark red (ischaemic) vs bright red (nonischaemic).

TABLE 73.2 CORPUS CAVERNOSAL BLOOD GAS PARAMETERS: ISCHAEMIC VERSUS NONISCHAEMIC PRIAPISM

Parameter	Ischaemic	Mixed venous (flaccid)	Nonischaemic
Colour	Dark red	Medium red	Bright red
pH	<7.25	7.35	7.4
PO ₂ (mm Hg)	<30	40	>90
PCO ₂ (mm Hg)	>60	50	<40

- The table places the flaccid penis ($\approx$ mixed venous blood) **between the two poles**: ischaemic is worse than venous; nonischaemic resembles **arterial** blood. **Trap**: expecting nonischaemic gas to look venous.

Adjunctive Testing

- Establishes aetiology; **performed in parallel, never delaying emergent treatment**, and unlikely to change acute management.
- **Complete blood count**: leukocytosis in leukaemia; also in sickle cell priapism, with **low haemoglobin** from haemolysis.
- **Haemoglobin electrophoresis** if haemoglobinopathy suspected (e.g. abnormal reticulocyte count) — **nonemergent**.
- **Coagulation profile**, especially with pharmacological anticoagulation.
- **Serum and urine toxicology**; drug exposure is often better elicited by history.

Imaging

- **Nonessential in routine acute diagnosis**; adjunct for specific scenarios — chiefly an **equivocal blood gas**, especially recurrence after recent treatment (recurrent ischaemic vs conversion to nonischaemic).

IMAGING MODALITIES COMPARED

Modality	Role	Findings / limitations
Penile duplex ultrasonography (PDUS)	Most accessible; point of care	Ischaemic: absent cavernosal flow, or peak systolic velocity (PSV) <50 cm/s with no diastolic flow . Nonischaemic: PSV >50 cm/s , may locate the fistula. Operator- and site-dependent; anatomy hard to read acutely. Post-intervention: documents reperfusion when examination and gas are equivocal
Magnetic resonance imaging (MRI)	Limited; nonacute	Tissue viability, necrosis, anatomy; poorly available acutely; needs subspecialist interpretation
Pelvic angiography	Not for diagnosis	Excellent for fistula and aneurysm, but invasive and costly; reserved for therapeutic embolisation

Figure 73.1 Algorithm for the diagnosis of priapism (modified from AUA/SMSNA 2022)

- **History + examination + corporal blood gas** → three outcomes:
- **Suggests ischaemic** (fully rigid, painful, pH <7.25, PCO₂ >60, PO₂ <30) → acute ischaemic management algorithm (Figure 73.2).
- **Indeterminate** → **penile and perineal duplex ultrasonography**.
- **Suggests nonischaemic** (partial rigidity, little or no pain, pH 7.4, PCO₂ <40, PO₂ >90) → nonischaemic algorithm (Figure 73.10).

Assigning Disposition

- **Ischaemic** → **emergent, expedient intervention**: necrosis, fibrosis and long-term ED are **time-dependent**; irreversible ischaemic damage begins **4–12 hours** after presentation.

- **Nonischaemic → nonemergent, expectant:** high inflow, adequate perfusion, minimal short-term tissue risk; **painless**, so no analgesia is needed; definitive treatment can be arranged non-urgently.
- **Stuttering → acute and chronic:** each episode is ischaemic and managed emergently; **pharmacological prophylaxis** is started to prevent further episodes.

Figure 73.2 Algorithm for the management of acute ischaemic priapism (modified from AUA/SMSNA 2022)

- Diagnose acute ischaemic priapism → **discuss risk of ED** → stratify by duration.
- **<36 hours:** goals are pain management and preservation of erectile function → **corporal aspiration ± irrigation + repeated intracavernosal phenylephrine**, monitoring heart rate and blood pressure → persistent → **distal shunt ± corporal tunnelling** → persistent after shunt without tunnelling → **corporal tunnelling** → persistent after shunt with tunnelling → further escalation.
- **>36 hours:** **counsel on high risk of ED** AND pain control with outpatient follow-up, OR **proceed directly to early penile prosthesis**.

KEY POINTS: DIAGNOSIS OF PRIAPISM

- The primary goal of diagnosis is to distinguish ischaemic versus nonischaemic priapism.
- The type of priapism can be determined through a combination of history, physical examination and corpus cavernosal blood gas assessment.
- Imaging is not essential for the routine diagnosis of priapism, but PDUS may be useful as an adjunctive diagnostic test in specific circumstances.

Concept — *Diagnosis is one question — is the corpus cavernosum ischaemic? — and corporal blood gas answers it; history and examination predict the answer, and PDUS rescues an equivocal result.*

MEDICAL THERAPIES

Ischaemic Priapism

- **Prompt, expeditious treatment;** delay risks permanent ED.
- Cold compresses, cold baths, exercise, induced ejaculation, cold-water enemas or oral medication **must never replace or delay first-line treatment**.

Conservative Therapies

- **Limited role;** must not delay definitive therapy.
- **Supportive care** (cold, ice, exercise) after oral or ICI ED drugs achieved detumescence in **only a minority** (retrospective series, incomplete follow-up).
- **Ateyah et al. (2005)** randomised controlled trial (RCT), ICI-induced priapism after failed cooling: aspiration plus irrigation at 10, 15, 20 or 37 °C — **cold 10 °C saline: 96% detumescence vs 60% at body temperature** → recommended **cold (10 °C) saline** irrigation.
- **SCD series:** IV hydration alone resolved some paediatric cases, others needed **exchange transfusion or shunting**; in Nigeria, conservative care (IV fluids, nonsteroidal anti-inflammatory drugs, adrenaline, stilboestrol) worked in a minority over days, the majority needing a **glanular-cavernous shunt**.
- Earlier Nigerian paediatric series: nearly all had SCD; **most responded conservatively**, surgical outcomes "poor". Later series: **Al-Ghorab shunt gave immediate detumescence** in almost all → timely diagnosis and surgery advocated.
- **AUA:** IV hydration and other conservative measures are **adjunctive only**.

Oral Medications

- **Not first-line** for acute ischaemic priapism (>4 hours). **First-line = corporal aspiration + ICI phenylephrine ± irrigation.**
- Historically oral vasoconstrictors were used; **few studies show efficacy.**
- Oral adrenergic agents studied: **pseudoephedrine** (α -agonist, analogous to phenylephrine), **terbutaline**, **phenylpropanolamine**, **ephedrine**.

ORAL AGENT TRIALS FOR PHARMACOLOGICALLY INDUCED PROLONGED ERECTION

Study	Setting	Comparison	Verdict
Three-arm placebo-controlled trial	After ICI PGE1	Terbutaline vs pseudoephedrine vs placebo	Terbutaline > pseudoephedrine > placebo
Double-blind RCT	After ICI	Terbutaline vs placebo	No benefit
Randomised study	Priapism	Oral terbutaline vs placebo	Benefit with terbutaline

- The trials **disagree**, which is why terbutaline carries **no AUA recommendation**. **Trap:** treating a positive single trial as grounds for first-line oral therapy.

Aspiration/Irrigation of Corpora Cavernosa

- Aspiration ↓ **intracavernosal pressure**, removes hypoxic, acidotic, hypercarbic blood and **relieves pain**.
- Technique: single large-bore **19-gauge needle on a 10-mL syringe** → enter the corpus cavernosum at the **penoscrotal junction at 3 or 9 o'clock** (avoids the dorsal neurovascular bundle) → aspirate while **compressing the shaft between thumb and index finger** until soft → adjust needle for return → leave needle in, let the shaft refill → recompress and re-aspirate → **repeat serially** → for irrigation, exchange the syringe for normal saline through the same needle.

ASPIRATION AND IRRIGATION OPTIONS COMPARED

Approach	Efficacy	Status
Aspiration alone	<1 in 4	Insufficient alone
+ saline (body temperature)	Relatively ineffective	Insufficient
+ cold 10 °C saline	High (Ateyah RCT)	Supported; avoid chilled saline in SCD
+ heparinised saline	Better than aspiration alone	Too few patients to recommend
+ intracavernosal phenylephrine	Standard	AUA first-line

Intracavernosal Injection (ICI) of Vasoactive Medication

- **AUA guideline: first-line = intracavernosal phenylephrine + corporal aspiration, ± irrigation.**
- **Sympathomimetics:** phenylephrine, etilefrine, ephedrine, adrenaline (epinephrine), noradrenaline (norepinephrine), metaraminol → **cavernosal smooth muscle contraction**.
- Toxicity: α → hypertension; β → inotropy/chronotropy; **most importantly reflex bradycardia**. Adverse effects: headache, hypertension, arrhythmia.
- **Blood pressure and cardiac monitoring for all**, especially with cardiovascular disease, hypertension, stroke or **monoamine oxidase inhibitors (MAOIs)**.
- α -agonists **contract cavernous smooth muscle** so trapped sinusoidal blood drains; β -agonists relax smooth muscle and dilate arteries, admitting oxygenated blood to wash out the sinusoids.
- Case-report efficacy of these agents is **variable**.

SYMPATHOMIMETIC RECEPTOR PROFILES

Agent	Receptor activity
Metaraminol	Pure α
Phenylephrine	Mixed, β effects minimal (relatively selective α_1)
Etilefrine, adrenaline	Mixed α and β
Terbutaline	β (β_2 with minor β_1)

- Technique: **penile block** with local anaesthetic at the base → 19-gauge needle into the **lateral shaft** → aspirate about **10 mL** to decompress → inject phenylephrine through the same needle.
- Dilute in normal saline to **100–500 mcg/mL**; give **1-mL aliquots about every 5 minutes** until resolution or futility; **continue up to 1 hour** (AUA).
- Expanded protocol: irrigate with normal saline (**chilled if available and the patient does not have SCD**) → aspirate blood and clots → inject phenylephrine → **observe 3–5 minutes** → repeat.

Concept — First-line therapy is **mechanical plus pharmacological**: aspiration empties the compartment and an α -agonist contracts the smooth muscle so it stays empty; everything conservative or oral is adjunctive.

Stuttering Priapism

- Most agents are **prophylactic**, for frequent and troublesome episodes; **evidence is largely anecdotal**.

Oral Alpha-Adrenergic Agents*Pseudoephedrine*

- Sympathomimetic of the **phenethylamine and amphetamine** classes; decongestant, also used as stimulant.
- Often used as **off-label first-line for stuttering priapism**; α -mediated corporal contraction.
- Dose poorly defined; **small numbers of SCD patients respond**.

Terbutaline

- **β_2 -agonist with minor β_1 effects**; postulated to alter cavernosal permeability and $\uparrow$ sinusoidal plasma flow.

Etilefrine

- **Selective α_1 -agonist** (possible β_1 chronotropic/inotropic effect); $\downarrow$ venous pooling, $\uparrow$ blood pressure; cavernosal contraction → detumescence.
- **Reduced episodes in most SCD patients** without altering baseline erectile function, but a **Cochrane systematic review found insufficient evidence**.

Hormonal Therapies*GnRH agonists*

- **Goserelin, leuprolide**: initial luteinising hormone (LH) and testosterone **flare** → receptor **desensitisation** → castrate LH and testosterone within weeks.
- Adverse effects: hot flushes, gynaecomastia, **loss of sexually induced erections**, $\downarrow$ spermatogenesis, asthenia, **loss of libido** → quality-of-life cost.
- Case: non-SCD man on adrenaline self-injection + monthly GnRH analogue → no recurrence, erectile function preserved; **cessation usually brings recurrence**.

Antiandrogens

- **Flutamide, bicalutamide, chlormadinone, cyproterone acetate**; relieve stuttering priapism **without the libido loss** typical of GnRH agonists and oestrogens.
- Bicalutamide effective in SCD and **spinal cord injury (SCI)**, with dosing frequency later reduced; chlormadinone **100 mg** → ED at 6 months, **recovered after withdrawal**.
- **Cyproterone**: ↓ LH, follicle-stimulating hormone (FSH) and testosterone; high complete response but **fatigue, hot flushes, breast tenderness, mood change, ED** → **high dropout**.
- Evidence is **small, non-randomised and marred by poor compliance**, hence no consistent improvement in severity or frequency.

5α-Reductase inhibitors

- Block **testosterone** → **dihydrotestosterone**; licensed for alopecia and benign prostatic hyperplasia (BPH); off-label for idiopathic and stuttering priapism.
- SCD series: **finasteride 5 mg (days 1–40) → 3 mg (days 41–80) → 1 mg (days 81–120)** → marked fall in monthly episodes.
- **3 and 5 mg optimal**; episodes rose on 1 mg; **gynaecomastia**; fertility effect in young men unassessed; RCT needed.

Other Agents

Digoxin

- Inhibits smooth muscle **Na⁺/K⁺-ATPase** → ↑ intracellular Ca²⁺ → **cavernosal contraction**; also impairs NO-mediated relaxation.
- May ↓ libido, excitement and rigidity, more at **0.25–0.5 mg/day**; a placebo-controlled trial showed **no fall in testosterone, oestrogen or LH**.
- Idiopathic stuttering: maintenance therapy → **fewer hospital visits, better quality of life**.

Baclofen

- Centrally acting **γ-aminobutyric acid (GABA) derivative** for spasticity; inhibits ejaculation; prevents **reflexogenic erections** in neurological disease and spinal cord lesions.
- **Intrathecal more effective than oral**: first intrathecal pump for recurrent priapism in SCI, refractory to oral 75 mg/day but responsive to a 25 µg intrathecal test dose.
- Oral **40 mg nightly** abolished idiopathic nocturnal recurrent priapism with normal erectile function.
- Adverse effects: drowsiness, nausea, fatigue, ED; **optimal prevention strategy unknown**.

Phosphodiesterase type 5 inhibitors (PDE5Is)

- **NO/cGMP dysregulation** is the primary molecular mechanism of recurrent ischaemic priapism; **PDE5 downregulation** → **cGMP accumulation** → **exaggerated erection**.
- Sildenafil, vardenafil, tadalafil, udenafil, avanafil: a **novel prophylactic option**.
- **Burnett et al. (2014) RCT, SCD**: sildenafil 25 → 50 mg daily vs placebo; **primary endpoint (≥50% reduction) not met** on intention-to-treat or per-protocol analysis; in the **open-label extension most reported ≥50% reduction**.
- **Tadalafil long-term**: alleviated recurrences and **improved erectile function** in idiopathic (post-Winter shunt) and SCD cases.

PROPHYLACTIC AGENTS FOR STUTTERING PRIAPISM

Agent	Class / target	Dose in text	Evidence	AUA stance	Signature toxicity
Pseudoephedrine	Oral α -agonist	60–120 mg nocte	Small case numbers	Off-label first-line in practice	Sympathomimetic
Terbutaline	β_2 -agonist	5–10 mg	Conflicting trials	Not recommended	—
Etilefrine	α_1 -agonist	50–100 mg daily	Small SCD series; Cochrane negative	—	β_1 cardiac effects
GnRH agonists	LH suppression	Monthly depot	Case reports	—	Loss of libido, gynaecomastia, hot flushes
Antiandrogens	Androgen receptor / LH	Flutamide 125–250 mg tds; bicalutamide 50 mg	Case reports; high dropout	No consistent benefit	Cyproterone: fatigue, ED, mood
Finasteride	5 α -reductase	5 $\rightarrow$ 3 $\rightarrow$ 1 mg taper	One SCD series	—	Gynaecomastia; fertility unknown
Digoxin	Na ⁺ /K ⁺ -ATPase	0.25–0.5 mg daily	Small cohort	Not mentioned	$\downarrow$ libido, rigidity
Baclofen	GABA-B, central	40 mg nocte oral; intrathecal	Case reports	Listed as preventative	Drowsiness, ED
PDE5Is	Restore PDE5	Sildenafil 25–50 mg daily; tadalafil 5–10 mg	Only RCT (Burnett 2014): primary endpoint missed	Most promising	—

- The table shows **no agent with robust evidence**; choice turns on side-effect burden. **Trap**: hormonal agents work by suppressing nocturnal erections — and therefore suppress sexual function too.

KEY POINTS: MEDICAL THERAPIES

- Oral medications are not recommended for the primary management of either acute ischaemic or nonischaemic priapism.
- Corpus cavernosal aspiration and intracavernosal injection of phenylephrine, with or without irrigation, are first-line treatments for ischaemic priapism.
- Oral PDE5 inhibitors are among the most promising therapies for preventing stuttering priapism episodes, though supporting evidence currently is limited.

Concept — Prophylaxis attacks either the **trigger (androgen-driven nocturnal erections)** or the **molecular imbalance (α -tone, PDE5)**; PDE5 inhibitors are the paradox that follows directly from the low-PDE5 model.

SURGICAL THERAPIES

- Indicated when **first-line measures fail**. Goals: **restore ingress and egress of blood**, restore homeostasis, prevent ischaemia and fibrosis.
- Most procedures create a **corpus cavernosum–corpus spongiosum fistula**; historical fistulae to the **dorsal penile or saphenous vein** have limited supporting data.

- All shunts are expected to **close spontaneously** after detumescence, with hope of return of erectile function.
- **Duration determines outcome:** longer priapism → higher chance of permanent ED; **>24 hours carries a very high ED rate.**

DURATION OF PRIAPISM AND ED — BENNETT AND MULHALL (2008), SCD

Duration	ED rate
≤12 hours	0%
12–24 hours	22%
24–36 hours	56%
≥36 hours	100% (complete erectile loss)

- This gradient is the **rationale for the 36-hour prosthesis threshold.**

Distal Percutaneous Corporoglanular Shunt Procedures

- After **failed phenylephrine and aspiration ± irrigation** → **consider a distal corporoglanular shunt (AUA).**
- **Optimal distal shunt undefined:** no head-to-head comparisons, and **no comparison of shunt alone vs shunt with tunnelling.**

Winter Shunt Procedure

- **Tru-Cut biopsy needle** through the glans into the corpus cavernosum → removes a small core of tunica between glans and cavernosum → fistula → **several passes bilaterally.**
- **Lower success than scalpel shunts** (Ebbehoj, Al-Ghorab, T-shunt); AUA still recognises it as valid.

Ebbehoj and T-Shunt Procedures

- **Ebbehoj: scalpel number 11, 12 or 15** through the glans into the cavernosum → cavernosum–spongiosum fistula → drainage into spongiosal venous system → glans skin sutured.
- **T-shunt** (Brant et al., 2009): **scalpel number 10, 11 or 12** inserted vertically through the glans, lateral to the meatus at a safe distance → advanced into cavernosum → **rotated 90° laterally** → T-shaped tunical window → unilateral first; **contralateral T-shunt** if erection persists or returns.

Distal Open Corporoglanular Shunt Procedures

Al-Ghorab Shunt Procedure

- Conscious sedation, spinal or general anaesthesia → **transverse incision 1 cm distal to the dorsal coronal ridge** → sharp dissection to the distal corpora → **excise the tips of the corpora** → large cavernosum–spongiosum fistula → milk out deoxygenated blood → close skin.
- Most outcome data aggregate multiple distal shunts; Al-Ghorab-specific series show **durable detumescence in almost all**, including late presenters.

Al-Ghorab Shunt Procedure With Tunnelling (Corporal Snake) Maneuver

- **Burnett and Pierorazio (2009):** general anaesthesia → urethral catheter → **2-cm transverse glans incision** → dissect to corporal tips → **2-0 stay sutures** → incise corpora → milk out blood → **7/8 Hegar dilator** through the corporotomy to the **proximal phallus** (breaks thrombus, drains stagnant blood) → further milking → close glans → gentle compressive dressing.

DISTAL SHUNT WITH TUNNELLING — OUTCOME SERIES

Study	Procedure	Bottom line
Zacharakis 2014	T-shunt + tunnelling	100% resolution if <24 h; only 55% at 24–48 h
Series	Modified Al-Ghorab + snake	Most detumesced despite prior failures; refractory → prosthesis
Series	Al-Ghorab + tunnelling	All detumesced after multi-day priapism
Brant 2009	T-shunt ± tunnelling	Flow restored in nearly all, even after failed shunts; pain resolved in all; most regained function

- Tunnelling rescues even prolonged and previously failed cases, but **success still falls with duration**.

DISTAL SHUNTS COMPARED

Shunt	Access	Instrument	Anaesthesia/setting	Distinguishing feature
Winter	Percutaneous, glans	Tru-Cut biopsy needle	Local, bedside	Least invasive; lowest success
Ebbehoj	Percutaneous, glans	Scalpel 11/12/15	Local, bedside	Scalpel stab fistula
T-shunt	Percutaneous, glans, lateral to meatus	Scalpel 10/11/12, rotated 90°	Local, bedside	Larger fistula ; bilateral if needed
Al-Ghorab	Open , dorsal glans incision	Scalpel excision of tips	Sedation, spinal or general	Excises corporal tips
Corporal snake	Open (with Al-Ghorab) or via T-shunt	7/8 Hegar dilator	General	Tunnels proximally, breaks thrombus

Proximal Shunt Procedures**Open Corporospongiosal Shunt Procedures (Quackels Shunt, Sacher Shunt)**

- Both need **general anaesthesia** and a **transscrotal or transperineal** approach; bilateral shunts **staggered ≥1 cm** to enhance drainage and **reduce urethral stricture**.
- Longitudinal scrotal or perineal incision → dissect to the **bulb of the spongiosum** and urethra → excise a **1-cm ellipse of tunica** from cavernosum and spongiosum → evacuate stagnant blood → anastomose with **5-0 polydioxanone (PDS)**.
- Few outcome data; unilateral vs bilateral not compared. **AUA: low utilisation; proximal shunt only after failure of more established measures, including distal shunting with tunnelling.**

Saphenous Vein Anastomotic Shunt Procedure (Grayhack Shunt)

- **Supine frog-leg** → incision at the **saphenofemoral junction, 3–4 cm below the inguinal ligament** → mobilise saphenous vein about 10 cm → second incision at the lateral penile base to the tunica → ligate distal saphenous → swing proximal end up → anastomose to tunica with **5-0 PDS**.

Deep Dorsal Vein Anastomotic Shunt Procedure (Barry Shunt)

- **4-cm longitudinal incision** at the penile base → identify superficial or deep dorsal vein → mobilise, ligate distally, transect → excise tunical ellipse sized to the vein → **tension-free anastomosis** with 5-0 PDS.

Open Penoscrotal Decompression Procedure

- Described by **Morey**: **glans-sparing**, rapid detumescence.
- **Transverse penoscrotal incision** → skin-hook retraction, isolate and incise each corpus, holding sutures → **proximal corporotomy** → **paediatric Yankauer suction** passed distally and proximally to evacuate clot → close corporotomy → repeat on the contralateral side → layered closure.

- **Snake tunnelling** can be added; approach permits **concurrent penile prosthesis**.
- Unilateral decompression had some recurrences, all resolved by bilateral; **bilateral upfront → no recurrences**; most regained spontaneous erections.

PROXIMAL AND VENOUS PROCEDURES COMPARED

Procedure	Fistula / drainage	Approach	Key technical point
Quackels	Cavernosum → spongiosum, unilateral	Transscrotal/transperineal	1-cm tunical ellipses at the bulb
Sacher	Cavernosum → spongiosum, bilateral	Transscrotal/transperineal	Stagger ≥1 cm (urethral stricture)
Grayhack	Cavernosum → saphenous vein	Groin + penile base	SFJ 3–4 cm below inguinal ligament
Barry	Cavernosum → dorsal vein	Penile base, 4 cm	Tension-free 5-0 PDS
Penoscrotal decompression (Morey)	No fistula — evacuation	Penoscrotal	Yankauer; glans-sparing; allows prosthesis

- SFJ = saphenofemoral junction. Pattern: **further from the glans = more invasive and less supported**. **Trap:** offering a proximal shunt before distal shunting with tunnelling.

Consent

- **Complications of shunting:** penile oedema, haematoma, **urethral injury**, urethral fistula, **penile necrosis**, pulmonary embolism, infection.
- Counsel that **priapism itself causes ED**, and longer duration means greater risk.

Concurrent Penile Prosthesis Placement at the Time of Shunt Procedure

- Shunts are not always successful; **each hour and each procedure adds smooth muscle damage** → necrosis, fibrosis, ED, sometimes ↓ sensation.
- Early implantation **avoids fibrosis**; matured fibrosis makes dilation difficult or impossible, forcing aggressive dilation with **cavernosal or urethral perforation**.
- Benefits of early placement: **preserved length**, easier insertion, shorter operating time, fewer reoperations.

EARLY VS DELAYED PROSTHESIS — ZACHARAKIS ET AL. (2014)

Outcome	Early	Delayed
Revision	9%	27%
Penile shortening	3%	40%
Rigidity for intercourse	High	High
Satisfaction	96%	60%
Erosion	0%	4%
Fibrosis at surgery	Minor distal in few	Dense in all
Infection	7%	19%
Malfunction	0%	4%

Outcome	Early	Delayed
Curvature	Rare	None

- Almost every outcome favours early placement; **shortening drives the lower satisfaction** after delay. **Trap:** rigidity is equal — the difference is length and complications.
- **AUA/SMSNA:** clinicians **may consider prosthesis for untreated acute ischaemic priapism >36 hours, or refractory to shunting ± tunnelling.**
- **Distal corporal perforation in up to 6%** after prior distal shunts; if erectile recovery is unlikely, weigh the **detriment of distal shunting** against a future implant.
- Other factors: **infection risk from repeated bedside irrigation**, corporal integrity after shunts, and **lack of optimisation** (poorly controlled diabetes, anticoagulants).

KEY POINTS: SURGICAL THERAPIES

- Because priapism duration is highly correlated with ED risk, surgical treatment should be promptly performed when conservative measures fail.
- Distal shunt (Winter, Ebbehøj, T-shunt and Al-Ghorab) procedures create corpus cavernosal–corpus spongiosum fistulae to facilitate penile detumescence.
- The penoscrotal decompression procedure is performed via a penoscrotal incision, using a proximal corporotomy to achieve detumescence.
- Corpus cavernosal snake ("tunnelling") manoeuvres can be used with proximal or distal shunt procedures to facilitate sustained detumescence.
- Patients with prolonged ischaemic priapism and a high risk for ED should be considered for early penile prosthesis placement.
- Early prosthesis placement has better outcomes and greater patient satisfaction than delayed placement.

Concept — Every shunt is **the same operation at a different address** — a controlled fistula escalating distal → tunnelling → proximal as invasiveness rises and success falls; **duration, not the choice of shunt, sets erectile prognosis**, which is why prosthesis enters at 36 hours.

MANAGEMENT OF HIGH-FLOW (ARTERIAL) PRIAPISM

- Persistent erection from **unregulated cavernosal arterial inflow**; first described by **Burt et al. (1960)**.
- **Boys and men <55 years: trauma** — the mechanisms listed under aetiology, plus kicks, **birth-canal trauma in newborns**, complications of penile diagnostics and surgery.
- **Men >55 years: often advanced malignancy eroding the cavernosal artery.**
- Most fistulae arise from the **cavernosal artery or its branches**.
- Haemodynamics: laceration → direct, persistent, irregular flow into the lacunae → high-velocity turbulent flow → **shear stress** → **endothelial NO release** → arterial and trabecular dilatation.
- **Incomplete rigidity** reflects **incomplete smooth muscle relaxation**.

Expectant Management

- **Not a medical emergency**; no ischaemia → **conservative management first-line**; **about 60% resolve spontaneously**.
- **Penile compression, ice, observation** → cavernosal artery vasospasm → clot → fistula closure.

- Non-responders: **colour DUS for fistula size and location** → informs the interventional radiologist on isolation and access route; allows longitudinal tracking.

Figure 73.10 Algorithm for the management of nonischaemic priapism (modified from AUA/SMSNA 2022)

- Diagnose nonischaemic priapism → **observation and expectant management, reassess at 4 weeks** → resolves, or persists.
- Persists → **penile and perineal duplex Doppler** (fistula presence, location, size) → discuss risks and benefits of **interventional radiology embolisation**.
- Patient opts for treatment → **cavernosal artery embolisation** → persists → **repeat embolisation**, considering **non-absorbable material if absorbable was used first**. Patient declines → continued observation and reassessment.

Interventional Radiology Angiography and Embolisation

Selective Pudendal Arterial Catheterisation With Embolisation

- **Five agents**, temporary or permanent (table); resolution in $\geq 75\%$.
- **Recurrence historically 30–40%** → repeat procedures often needed.
- **AUA**: recurrence after embolisation → **repeat embolisation, not open ligation**.

EMBOLISATION AGENTS COMPARED

Agent	Type	Distinguishing feature
Autologous clot	Temporary	Used in children ; recurrence within hours is frequent
Gelatin sponge	Temporary, 2–6 weeks	Detumescence without permanent ablation ; reabsorbed
Microcoils	Permanent	Theoretical $\uparrow$ risk of permanent occlusion → ED
Polyvinyl alcohol (PVA) particles	Permanent (nonresorbable)	Preferred for repeat embolisation (AUA)
N-butyl cyanoacrylate glue	Permanent, ionic	Polymerises on blood contact; second line for recurrence ; ED risk $\downarrow$ by direct puncture of the pseudoaneurysm

- The pair tested: **temporary (recurrence) vs permanent (theoretical ED)**. Recent series report **lower long-term ED with permanent agents**, reversing the historical view.
- **Trap**: after a failed resorbable embolisation, the next step is **repeat embolisation with a nonresorbable agent**, not surgery.

Transcorporal Selective Surgical Ligation of Fistula in Select Cases

- Reserved for **failure of repeated embolisation**.
- **Lateral inguinoscrotal incision** → expose proximal dorsal neurovascular bundle → identify cavernosal and distal common penile arteries → **intraoperative PDUS** localises the fistula (compression $\downarrow$ velocity confirms) → small corporotomy → **ligate and transect** → detumescence.
- **Higher ED risk** than other treatments (about half); complications: ED, **penile gangrene**, gluteal ischaemia, purulent cavernositis, perineal abscess.
- **AUA/SMSNA**: after failed first embolisation, offer a **second, preferably with nonresorbable PVA particles**, especially if the first used a resorbable agent; repeat embolisation is safer and more effective than surgery, whose benefit cannot be quantified.

KEY POINTS: INTERVENTIONAL RADIOLOGY

- Selective cavernosal arterial embolisation has high success rates in treating nonischaemic priapism that persists after expectant management.
- Repeat embolisation is generally preferred over open surgical fistula closure if initial embolisation is unsuccessful.

Concept — *Because the corpora are well oxygenated, **time is not tissue in high-flow priapism**; management escalates by invasiveness, trading each step's erectile risk against persistence.*

SUMMARY

- **History, examination and corporal blood gas** remain the diagnostic cornerstone; delay to resolution drives fibrosis, shortening and ED.
- Ischaemic ladder: **phenylephrine + aspiration ± irrigation** → distal shunt ± tunnelling → proximal shunts (limited success); **open penoscrotal decompression ± tunnelling** is the newer proximal option with promising results.
- Stuttering: episodes **typically ≤1 hour**; pharmacological prevention, plus **self-administered intracavernosal phenylephrine at home** for episodes after onset.
- Nonischaemic: expectant management resolves most; **angiographic embolisation** next.
- **Research needs**: molecular mechanisms of ischaemic and stuttering priapism; outcomes of **concurrent vs delayed prosthesis**.